APRIL 1, 2005

INORGANIC CUMULATIVE EXAMINATION

1. Allen Bard (U of T, Austin) recently reported the following cyclic voltammograms of O_2 in liquid ammonia.

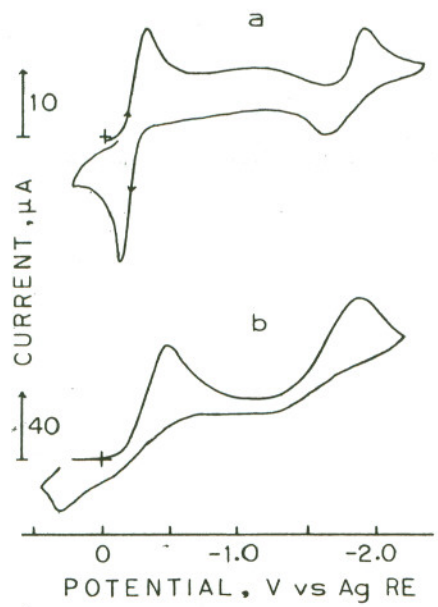

Figure 2. Cyclic voltammograms of O_2 (1 atm total pressure) in liquid NH_3 saturated with $MeBu_3NI$ (−55 °C): (a) at Pt disk electrode (0.039 cm^2), 50 mV/s; (b) at glassy carbon electrode (0.071 cm^2), 200 mV/s.

Rationalize the number and reversibility of the observed peaks as well as the differences noted with the different electrode materials.

2. What are the advantages of using NH_3 in these experiments as opposed to water? Your discussion should include specific chemical reactions.

3. With the aid of a M.O. diagram, rank the species O_2^+, O_2, O_2^-, and $O_2^=$ in order of increasing O-O stretching frequency. What are singlet and triplet oxygen (use M.O. diagram)? Where would they fit in this ranking?